

SAVITHA VISWANADH KANDALA

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RESEARCH FOCUS

Low-power systems and networks for embedded devices, with a focus on multi-microcontroller platforms, backscatter communication, and edge AI.

EDUCATION

Ph.D. in Computer Science,
National University of Singapore (NUS)
Advisor: **Dr. Ambuj Varshney**

Aug 2024 – Present

B.Tech. and M.S. by Research in Electronics and Communication Engineering,
International Institute of Information Technology, Hyderabad (IIIT-H)
CGPA: 9.01/10.0

2019–2024

RESEARCH EXPERIENCE

National University of Singapore
Graduate Researcher

Aug 2024 - Present

- MicroSymphony** – Compute-first multi-microcontroller platform for concurrent communication, sensing, and inference at microwatt power
 - Designed a scalable 4 MCU MSP430 testbed enabling concurrent distributed execution across controllers
 - Built shared-FRAM IPC and multi-drop UART fabric with near bus-limited throughput and < 2% overhead
 - Developed a custom bootloader and flashing protocol for > 10× faster multi-MCU firmware programming
 - Enabled distributed on-device ML inference with ~ 65% lower latency and 35x lower sampling jitter
 - Released hardware, firmware, and toolchain as open-source
- AutoTag** – Realizing the decade-old vision of long-range, multi-hop tag-to-tag backscatter networks
 - Demonstrated 70 m tag-to-tag links (14× SoTA) with FDM-based concurrent transmissions
 - Developed a low-power multi-channel PHY and MAC protocol for networking of tags
 - Observed over 100s of meters communication in non-line-of-sight simulations using signal-gain mechanisms
- TinyLLM** – Framework for training and deploying foundational language models at the edge
 - Pre-trained custom 30–124 M-parameter SLMs for sensor and IoT analytics on H100s
 - Curated a 10 B-token web/code/sensor corpus enabling task-specific edge pre-training
 - Released **open-source models** matching Phi-3/Llama-3 on IoT tasks at <1 GB memory footprint
 - LoRA fine-tuning with 440 examples enables hand-gesture recognition; 2000+ GPU-hours total
 - Awarded the **2026 Google ML Research Award with TPUs** – TPU compute and Google mentorship

National University of Singapore
Visiting Scholar

Jan 2024 - Jun 2024

- Benchmarked LLMs (Phi, Gemma, Llama) across a range of single-board computers
- Evaluated fine-tuned LLMs for wireless sensing and sensor datasheet interpretation
- Implemented distributed inference of the Llama model across multiple single-board computers

- Developed **RLabs**, a low-cost remote labs solution featuring 5+ hardware experiments
- Collaborated with 6 professors and led a team of 5 research students and 15 interns
- Deployed with Agastya (NGO) to educate over 300 students across rural India
- Performed a security analysis on a large-scale **IoT-based air pollution monitoring deployment**, identifying and mitigating multiple threats

OTHER EXPERIENCES

Google

PhD Software Engineering Intern

May 2026 - Sept 2026

- Worked with the XProf team (CoreML) to improve hardware profilers for PyTorch on Google Cloud TPUs
- Built C++ layer attribution to improve host-device correlation for LLM training in XProf traces
- Analyzed communication-computation overlap across distributed TPU workloads to quantify network stalls
- Achieved up to 1.5x faster training via sub-layer FSDP un-sharding to hide communication behind compute

Walmart Global Tech India

Software Developer Intern

May 2023 - July 2023

- Built a retail object detection pipeline using RetinaNet for localization and Vision Transformer embeddings indexed with FAISS for fast similarity search across the product catalog
- Recognised 30+ shelf items per image with $\geq 95\%$ accuracy on Walmart’s internal benchmark

Teaching Assistantships

- *National University of Singapore* (Spring 2025, 2026) – Wireless Communications
- *IIIT-Hyderabad* (2021–2023) – Embedded Systems Workshop, Electronics Workshop, Statistical Methods in AI, Value Education, Communications & Controls in IoT

SKILLS

Programming	C/C++, Python, Bash, MATLAB, Java, HTML, JavaScript
Tools	Linux, Git, Slurm, Docker, L ^A T _E X, Wireshark, PyTorch, HuggingFace
Hardware	MSP430, ESP, Arduino, Raspberry Pi, Lattepanda, Orange Pi, MilkV, LuckFox

AWARDS & ACHIEVEMENTS

- ACM MobiCom 2024 Travel Grant
- NUS PhD Research Scholarship Awardee, 2024
- TIH-IoT CHANAKYA Fellow, 2022–2023
- Institute’s Best All-Rounder Award, 2023 (IIIT-H)
- IIIT-H Dean’s List (2019–2023)
- IIIT-H Research List (2021–2022)
- Best Poster Award at **IIIT-H’s R&D Showcase 2022**
- Solved ‘**IBM Ponder This**’ (8 times): Jan 2026; Nov, Dec 2025; July, Aug, Sept 2023; May, July 2022

ACTIVITIES

- Head, Electronics and Robotics Club & NSS Chapter, IIIT-H
- Coordinator, Student Placement Committee, IIIT-H
- Secretary, Mess Committee, IIIT-H
- Volunteer at Asha Kiran, an organisation educating underprivileged students